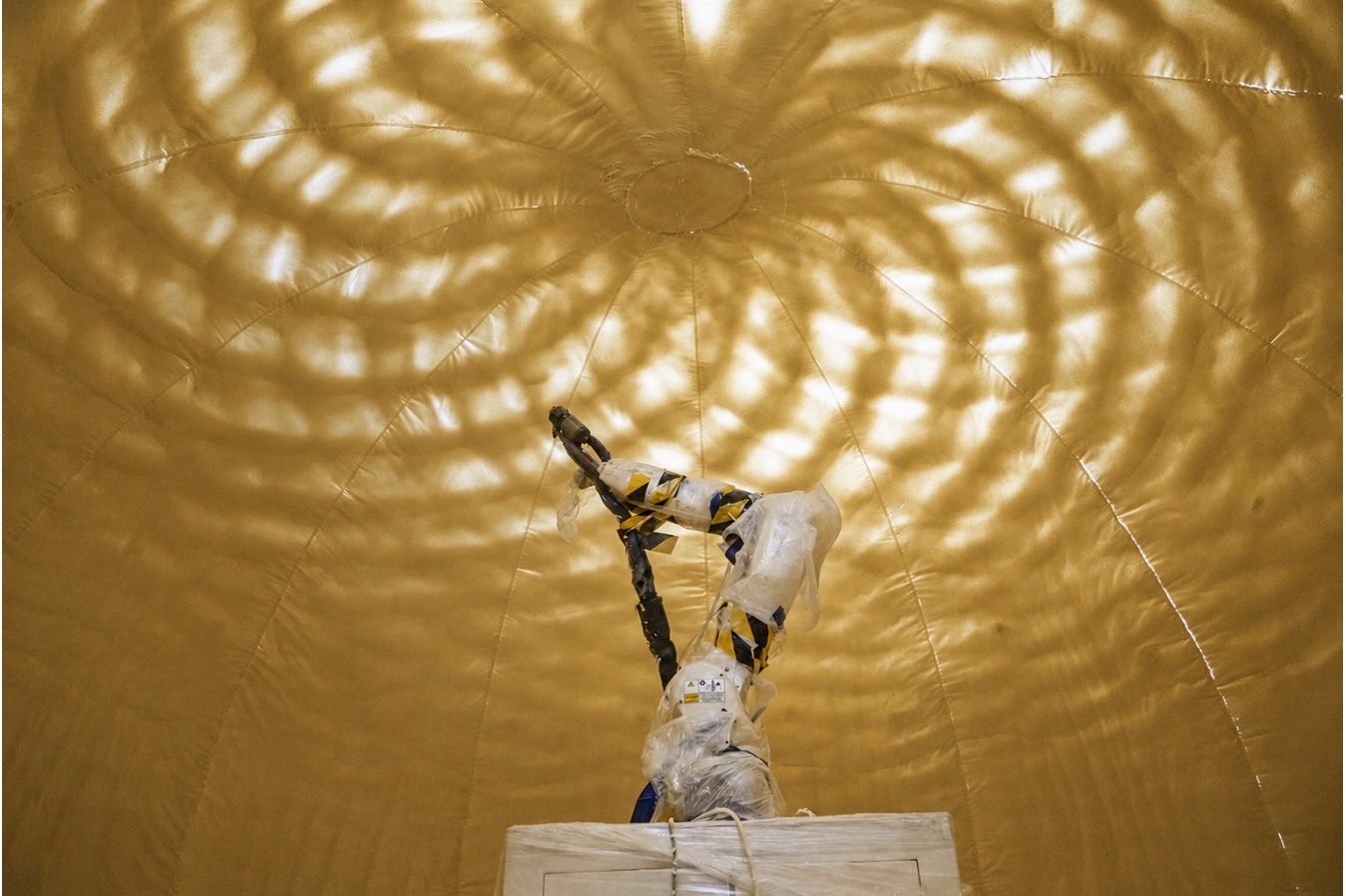


Minimum Device, Maximum Space

Reach-Aware Robotic Spray Fabrication of Inflatable Architectural Enclosures



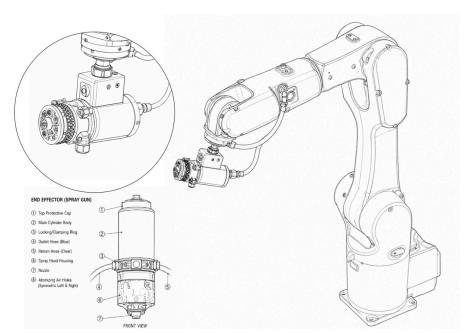
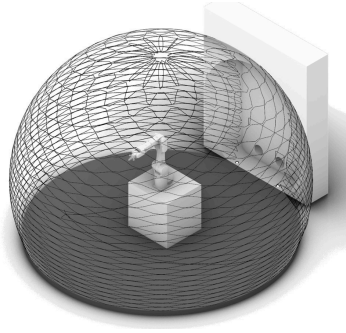
1 Inside-out robotic polyurethane spraying within an inflated membrane. A compact robot is positioned inside the pneumatic enclosure and projects PU onto the inner membrane surface, where the material is received, foamed, adhered, and cured into a continuous coating.

Keywords: robotic fabrication, inflatable formwork, polyurethane spraying, material process, prototype

SCALE BY INFLATION

Architectural-scale additive construction depends on the machine envelope, process planning, and material delivery as well as geometry (Paolini et al. 2019). Full-scale systems enlarge fabrication equipment, as in multi-story gantries, or mobilize robotic platforms across the site (Dörfler et al. 2022). Flexible formwork defines curved construction geometries before rigid material is added (Hawkins et al. 2016). Pneumatic systems generate these forms through air-supported membranes (Kromoser and Huber 2016), including reusable formwork for complex concrete shells (Boutemy et al. 2023).

This project uses inflation to generate the fabrication volume (Figure 2): the robot remains compact, while the architectural volume is produced first through inflation. Polyurethane is projected as a concentrated sprayed jet from a compact robot positioned inside the inflated membrane. The material travels through the interior air volume, lands on the PVC receiving surface, and consolidates through foaming, adhesion, and curing. This process forms a lightweight, rigid, large-span structure (Figure 1).



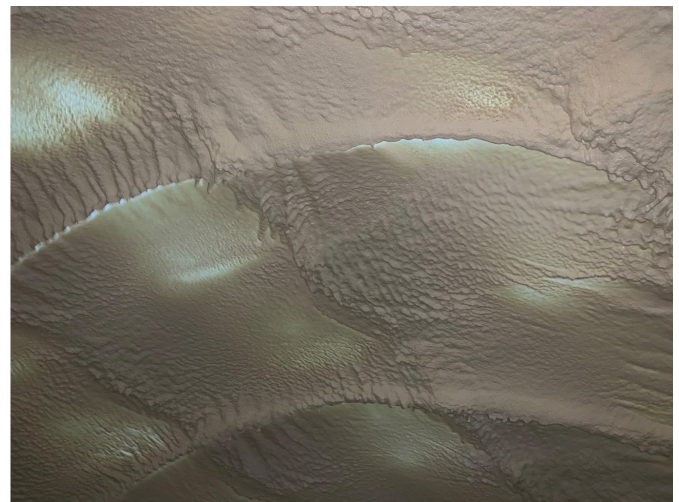
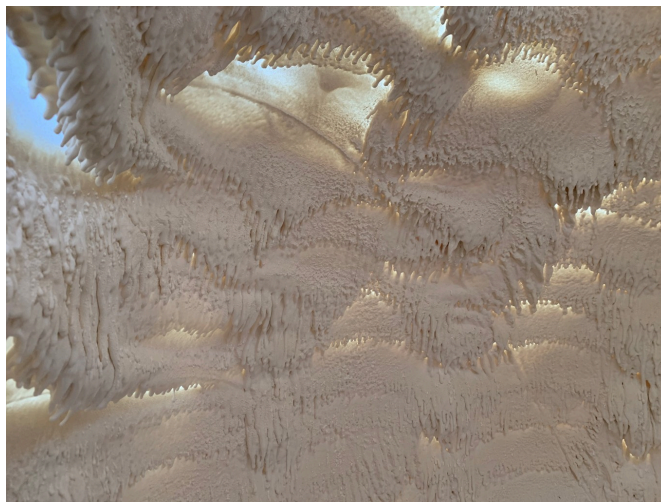
2 Overall method for inside-out robotic polyurethane spraying. (Left) Built sprayed membrane prototype with the robot positioned inside the PU-coated pneumatic enclosure. (Middle) Digital setup showing the inflated membrane, robot placement, and internal spray workspace. (Right) Robotic setup and custom PU spray end-effector used to project material onto the inner membrane surface.

INSIDE-OUT SPRAYING

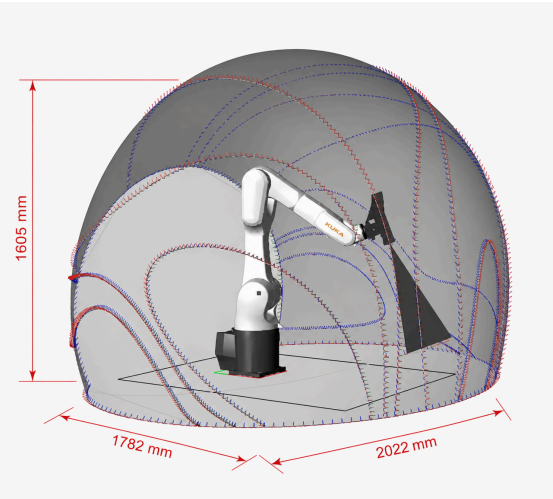
The method places a compact robotic arm inside an inflated membrane and uses it to spray polyurethane onto the PVC membrane enclosure from within. Spray-based fabrication depends on projection distance and overlap [Neudecker et al. 2016]. Process timing and receiving-surface behavior also govern deposition [Nuh et al. 2022]. Here the membrane acts as a pneumatic catcher: it sets the geometry, receives the projected PU, and temporarily supports it while the material expands, bonds, and stiffens. Deposition sequence therefore matters as much as toolpath geometry.

CALIBRATION THROUGH SMALL MEMBRANES

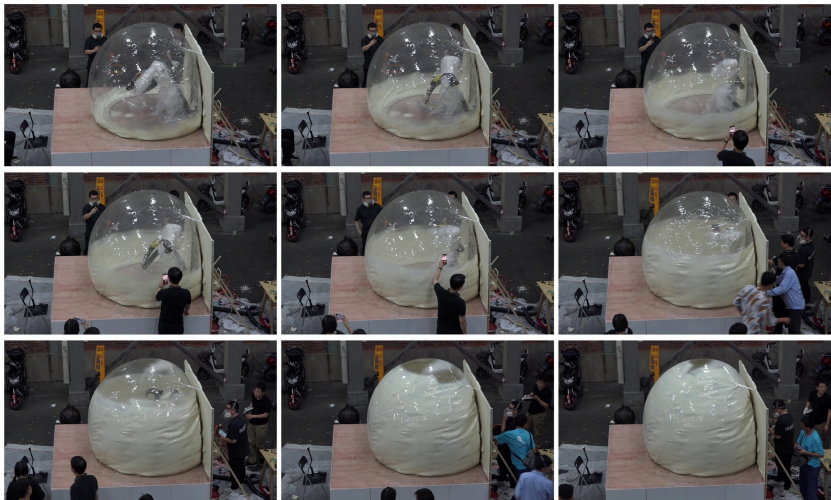
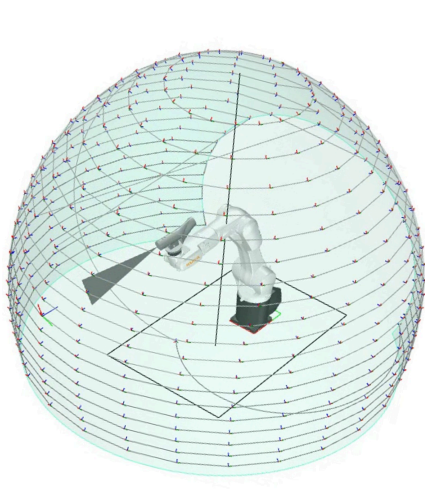
Early tests [Figure 4] used 2 m spherical membranes to define measurable inputs for robotic polyurethane spraying before transfer to the 5 m prototype. Six standoff settings from 20-70 cm produced deposited path widths of roughly 12-24 cm, which converted nozzle distance into a planning variable for band coverage on curved receiving surfaces. Substrate trials, as displayed in Figure 3, compared PVC, Oxford cloth, and carbon-fiber membrane because wet polyurethane had to remain attached during foaming, expansion, and curing. The reach study defined usable workspace through the overlap of robot motion, nozzle orientation, projected jet distance, and the portion of the membrane able to receive material [Figure 4]. The 2 m tests carried three variables into the 5 m prototype: projection distance, membrane retention behavior, and the usable spray workspace between the robot and the membrane.



3 Comparison of PU deposition behavior on two membrane substrates. Left: Oxford-cloth membrane, where the uncured PU showed poor early-stage retention, resulting in discontinuous buildup, dripping, and exposed gaps between spray passes. Right: PVC membrane, where the textured surface retained the uncured PU more effectively and produced a more continuous coating across adjacent robotic spray paths.



4 Small-scale spherical (two-meter span) PVC membrane geometry used for empirical reach calibration of the robotic PU spraying setup. The figure shows the robot positioned inside the membrane and identifies the projected spans (1782 mm and 2022 mm) and overall height (1605 mm) of the current test case. This geometry was evaluated together with minimum and maximum boundary cases in the digital model to determine the workable nozzle clearance required for PU deposition inside the pneumatic formwork.

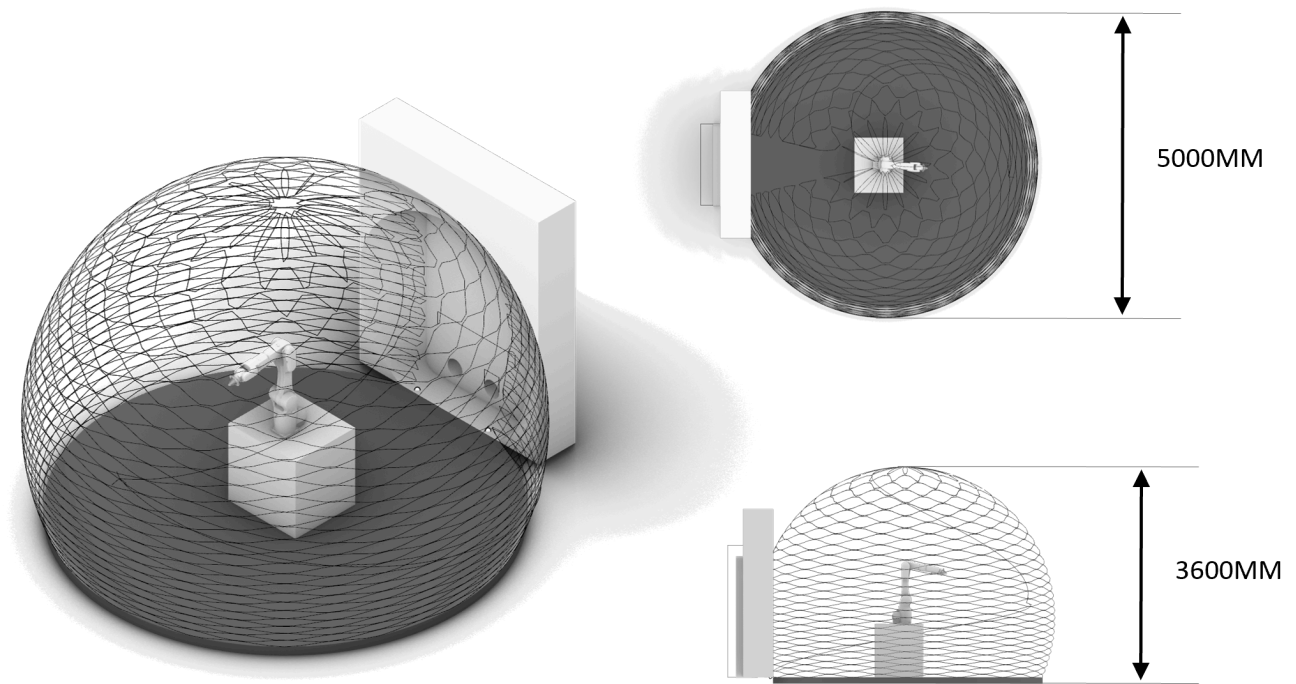


5 Sequential documentation of the bottom-up PU spraying process inside a small spherical PVC inflatable membrane. The robot deposited polyurethane on the inner membrane surface in staged bands, beginning near the lower perimeter and progressing upward. This deposition order reduced wet-material accumulation at the crown, limited membrane sagging before curing, and allowed previously sprayed lower regions to stiffen before additional PU was applied above.

5-M-SPAN PROJECTED DEPOSITION

The 5 m prototype (Figure 6) transferred the 2 m calibration rules to a membrane geometry that the robot could not surface-follow. In many regions, the polyurethane traveled more than 2 m through the inflated interior before contacting the membrane.

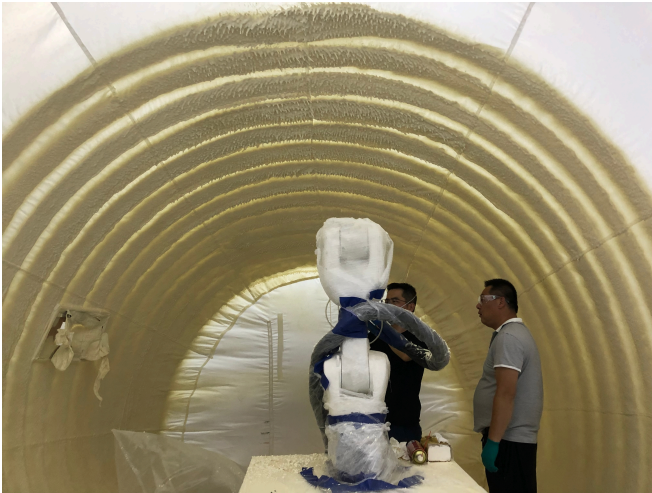
Deposition depended on directed material projection: nozzle orientation and robot speed controlled the impact area, while membrane curvature and surface condition controlled how wet polyurethane was received, held, and cured.



6 5 m-span inflatable membrane spraying setup. The robot is positioned inside the pneumatic enclosure, and the isometric, plan, and sectional views show the limited reach envelope, nozzle access, and spatial constraints that shaped the architectural-scale transfer.

The spray path was organized as staged bands beginning near the lower perimeter and progressing upward, as shown in Figure 6. This order kept the earliest wet material away from the crown, where gravity-driven flow and membrane deflection would amplify local accumulation before curing. Lower bands also had time to foam, adhere, and stiffen before additional polyurethane was applied above them. The sequence, therefore, used curing time as a temporary stabilization mechanism for a compliant receiving surface (Figure 7 and Figure 8).

At 5 m scale, a toolpath defined an impact zone shaped by projection distance, jet spread, robot speed, membrane angle, and polyurethane reaction state. Overspray remained inside the enclosure and produced secondary deposition: some material formed a rough texture, while some reattached outside the intended spray area. Sagging occurred when wet polyurethane accumulated faster than it could foam, adhere, and set. The hose drag limited wrist motion because the material lines added load and changed the robot posture during spraying. Operators adjusted nozzle direction, dwell time, and band sequence in real time because membrane deformation and curing state changed during deposition.



7 Interior view of the 5 m inflatable membrane during an early spraying test. The first lower-to-upper arched toolpath pass has been completed, leaving thick PU bands on the inner membrane surface while researchers inspect the curing state and local stiffness of the deposited material.



8 Exterior view of the 5 m inflatable membrane after internal robotic PU spraying. The PVC membrane has no external three-dimensional texture, but the internally deposited PU bands change local thickness and light transmission, producing a visible diamond-like pattern under interior illumination.

CONTRIBUTION

This project contributes a scale-transfer workflow for robotic spraying in which inflation supplies architectural span and a compact robot supplies internal materialization. The 5 m prototype identifies projected PU deposition as the key fabrication mechanism: the material travels through the inflated interior, contacts the PVC membrane, and stiffens through foaming, adhesion, and curing. The staged lower-to-upper toolpath turns curing time into a process constraint, because lower bands gain stiffness before later bands add material above them and the crown avoids early wet accumulation.

Mobile fabrication extends reach through robot repositioning [Gifftthaler et al. 2017], supported by sensing and in-situ coordination [Buchli et al. 2018]. This workflow extends fabrication scale through pneumatic pre-forming and material projection from a fixed internal robot. The project defines the next technical requirements for this method: controlled projection distance, band overlap, membrane retention, hose management, and real-time adjustment during curing.

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